

Multi-Label Intent Classifier for Google Play Store App Reviews

Rishika Vaish

The University of Texas at Dallas

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Abstract

This project addresses the growing need to derive actionable insights from the vast pool of user-generated app reviews. Traditional sentiment analysis offers a limited perspective, often classifying feedback into a single category. However, real-world user feedback is rarely that simple — it often includes a mix of praise, complaints, and suggestions. This project introduces a novel approach for multi-label intent classification combined with sentiment analysis to understand app reviews more comprehensively. Using a combination of TF-IDF clustering for label discovery and fine-tuned XLNet for classification, the model effectively identifies overlapping intents within each review. A responsive Streamlit application allows users to analyze reviews in real time. The goal is to empower developers and analysts with a tool that not only classifies sentiment but also pinpoints specific areas of concern or appreciation.

Problem Statement

Mobile applications rely heavily on user reviews to identify bugs, understand user needs, and prioritize feature updates. However, most tools today either perform basic sentiment analysis or assume that a single label can represent an entire review. This simplification leads to a loss of valuable information. For instance, a user might simultaneously report a failed recharge attempt, a lack of customer support, and an overall negative experience — all in one review. Thus, there is a critical need for a solution that can capture multiple user intents per review along with sentiment intensity, while being accessible through an intuitive interface.

Approach & Methodology

1. Data Collection

Reviews were programmatically extracted from the Google Play Store using the google-play-scraper package. The focus was on banking applications, which often attract a variety of user sentiments and complaints. This ensured diversity and relevance in the dataset.

2. Preprocessing

Raw reviews were cleaned and standardized using NLTK. The preprocessing steps included:

- Removing special characters, HTML tags, and excess whitespace
- Tokenization and stopword removal
- Lemmatization reduces words to their base form

This preprocessing ensured that the input to the model was consistent and semantically meaningful.

3. Sentiment Analysis

To capture the emotional tone of each review, the VADER sentiment analysis tool was used. VADER is specifically tuned for social media and short-form text, making it ideal for app reviews. It returns a compound score, which is then mapped to positive, neutral, or negative sentiment classes.

4. Intent Discovery

Rather than defining labels upfront, unsupervised learning was used to discover common themes:

- TF-IDF vectorization was applied to transform reviews into feature vectors
- K-Means clustering grouped reviews into conceptually similar clusters

- Each cluster was manually examined and assigned one of the following intent labels:
 - Problem with recharge
 - Problem with reward/redeem points
 - Problem in customer care service
 - Problem in registration/login
 - Other complaints
 - Appreciation
 - Bad/Irrelevant comments

This approach helped ensure that the labels reflected real user concerns.

5. Multi-Label Classification

A powerful transformer model — XLNet — was fine-tuned on the clustered and labeled data. Unlike traditional classifiers, this model is designed to handle multiple labels for a single input. The output is a set of binary predictions for each label, each accompanied by a probability score.

6. Web Application Deployment

To make the model user-friendly, it was deployed using Streamlit. The web interface:

- Accepts review input from the user
- Displays predicted sentiment and intent labels with confidence scores
- Is styled for clarity and responsiveness, enabling real-time feedback for app developers

Experiments & Results

The model was tested on a labeled validation dataset comprising a variety of user reviews. Evaluation metrics included precision, recall, and F1-score for each intent label. Results showed:

- F1-scores above 85% for clearly defined intents like "Appreciation" and "Recharge Issues"
- Robust handling of mixed-intent reviews
- Sentiment classification aligned closely with human interpretation

Several example reviews were analyzed to validate the model's predictions, demonstrating its ability to highlight nuanced user feedback effectively.

Testing with External Users

The prototype was tested by five users unfamiliar with the backend. The goal was to evaluate usability and prediction accuracy from an external perspective. Feedback highlights include:

-  Users found the interface intuitive and easy to use
-  Appreciated confidence scores and real-time feedback
-  Suggested improvements included reducing latency and adding a visual progress bar
-  One user recommended more descriptive intent names for clarity

Incorporating this feedback helped refine the user interface and improve backend response times.

Lessons Learned

This project reinforced several key lessons in both machine learning and user-centered design:

- **Performance vs. Usability:** Real-time deployment requires thoughtful optimization beyond just model accuracy.
- **XLNet's Versatility:** XLNet performed well in capturing multi-intent reviews compared to simpler architectures.
- **Clustering as a Bootstrap Tool:** TF-IDF + K-Means was effective for generating initial label sets, especially in the absence of manually labeled data.
- **Importance of Feedback Loops:** Testing with real users provided valuable insights that improved both model interpretation and interface design.

Future Work

Several avenues exist for improving this system:

- **Backend Optimization:** Use async processing or model caching to improve response speed
- **Progress Tracking:** Add a visual loading bar and log of past predictions
- **Multilingual Support:** Expand the classifier to support reviews written in other languages using multilingual transformers like mBERT or XLM-R
- **Gamification:** Introduce features like badges, streak counters, or feedback ratings to improve engagement in large-scale review analysis environments

Contributions

This was an individual project developed end-to-end by Rishika Vaish. Responsibilities included:

- Data scraping and cleaning
- Exploratory data analysis and clustering
- Model design and fine-tuning
- App development and deployment
- Presentation and documentation

Self-Scoring Table

Criteria	Points	Justification
Significant exploration	80	Implemented clustering, fine-tuned XLNet, and deployed a real-time application
Innovation or creativity	30	Used clustering for intent label generation, integrated multi-label NLP in production
Highlighted complexity	10	Built a full-stack ML system, handled real-world text and noisy data
Lessons learned + improvements	10	Documented iterative testing and optimization outcomes
Exceptional visuals/diagrams	10	Delivered well-structured PPT, diagrams, and a polished app interface
Testing with external users	10	Collected actionable feedback from five independent testers
Earned money	0	Not applicable

Total: 150 points

GitHub Repository

<https://github.com/rishika7006/nlp-project-multilabel-intent-classifier>

YouTube Video Presentation

https://youtu.be/BTbv_WKmj1g?si=ZQOHf2RdooR572V